

POKHARA ENGINEERING COLLEGE
Internal Assessment Examination

Level: Bachelor	Semester – Fall	Year :2025
Programme: B.E. Computer		Full Marks: 100
Course: Cloud Computing and Virtualization		Pass Marks: 45
Semester: 7th Semester		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Cloud Computing. Explain the need for cloud computing and describe its key characteristics as defined by NIST. 2+5
b) Explain the Cloud Reference Model. Describe the NIST CloudArchitecture. 2+6
2. a) Explain the evolution of cloud computing from traditional computing to modern cloud environments. 7
b) Explain Cloud Service Models (IaaS, PaaS, and SaaS). 8
3. a) Explain Virtual Private Cloud (VPC) in detail. Explain why VPC is required even though cloud resources are virtual. 2+5
b) Explain Virtualization and describe Type 1 and Type 2 Hypervisors with neat diagrams. 2+6
4. a) Explain Cloud Storage Models. Describe Object and BlockStorage with suitable examples. 2+5
b) Explain Database as a Service (DBaaS). Differentiate between Relational, Non-relational, and Graph databases used in cloud computing. 2+6
view data *chat* *relation*
5. a) Explain Resource Management in Cloud Computing. Describe vertical and horizontal scaling. 2+5
b) Explain Load Balancing in Cloud Computing. Describe any four load balancing algorithms and explain how high availability is achieved. 2+4+2
6. a) Explain Cloud Security and discuss major cloud security threats. 2+5
b) Explain Security as a Service (SECaaS). Describe cloud security 2+3+3

mechanisms such as IAM, and digital signatures.

2×5

7. Write short notes on any two:
 - a) Service Level Agreements
 - b) CSA Stack Model
 - c) Containerization

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Pokhara Engineering College
Internal Assessment

Level : Bachelor
Program : Computer (Sem 7)
Subject : Data Science and Analysis

Year: 2026
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe data preprocessing steps: data cleaning (missing values, outliers, duplicates) and data integration (schema/entity conflicts). Suggest suitable methods for each. 7
- b) Define variable and classify variables into numeric and categorical and their sub-types with examples. 8
2. a) Define statistics and explain how it supports data analysis in real-world domains 7
- b) A dataset ($n=10$) of study hours is: 2, 3, 3, 4, 4, 4, 5, 6, 6, 8.
(i) Construct the empirical frequency & relative frequency table.
(ii) Compute the Empirical Distribution Function (EDF) and interpret it. 8

OR

Given a dataset with X = number of study hours = [2, 3, 5, 6] and Y = marks = [50, 60, 80, 85]. Compute the Pearson's correlation and interpret your result.

3. a) Explain univariate and multi-variate data with suitable examples. 7
- b) A researcher is studying how study hours (X_1) and attendance rate (X_2) influence exam score (Y) of 5 students. The regression model fitted is: $\hat{Y} = 5 + 6X_1 + 0.3X_2$. Compute: SST, SSR, SSE, R^2 and Adjusted R^2 . The dataset is:

Student	X_1 (Hours)	X_2 (%)	Y (Score)
1	2	70	50
2	3	80	60
3	4	75	65
4	5	85	80
5	6	90	85

4. a) Derive the Nadaraya-Watson estimator for Non-parametric regression. Also, explain its significance. 7
- b) What is dimensionality reduction? Explain the steps involved in PCA analysis. 8
5. a) Explain permutation tests and partial dependence plots for testing the importance of variables in a dataset. 7

OR

$$\sigma = \frac{1}{1+e^{-z}}$$

Describe supervised and unsupervised learning. Explain the sigmoid function used in logistic regression.

- b) What is clustering? Generate cluster from following dataset using K-means algorithm (take $k=2$ and consider up to 2 iterations, use hamming distance). 8
 - T1: Bread, Jelly, Butter
 - T2: Bread, Butter
 - T3: Bread, Milk, Butter
 - T4: Coke, Bread
 - T5: Coke, Milk

6. a) What do you mean by times series data? Explain the components in a times series data. 7

- b) Consider a simple time series with 5 observations as below. 8
Compute autocorrelation at lag-1. Interpret out result.

Time (t)	Value X_t
1	2
2	4
3	6
4	8
5	10

7. Write short notes on (any two):

10

- a) Data Analysis
- b) Multi-collinearity
- c) Causation Vs Correlation

POKHARA ENGINEERING COLLEGE

Level: Bachelor
Programme: BE, Computer 7th

Year : 2025
Full Marks : 100
Pass Mark : 45

Course: Entrepreneurship and Professional Practice Time : 3 hrs

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. "A self-employed person is necessarily an entrepreneur". Do you agree with this statement? Defend your decision. 7
b. "Entrepreneurs are the backbone of economic growth." Justify this statement by explaining the importance of entrepreneurship. 8
2. a. What is innovation? Explain different types of innovation in entrepreneurship. 7
b. Discuss the various sources of startup funding. 8
3. a. Describe the entrepreneurship ecosystem and explain its key components. 7
b. What is competitor analysis? Explain different sources of competitive advantage. 8
4. a. Discuss, with examples, three key components of branding and three key components of promotional strategies before launching a mobile application 7
b. Define whistle blowing. Discuss its importance and ethical challenges faced by whistleblowers. 8
5. a. Explain the Code of Conduct of the Nepal Engineering Council (NEC) and its importance in professional practice. 7
b. Explain the Lean Canvas and describe its key building blocks. 8
6. a. What is meant by business plan? Explain its importance in startups. 7
b. Describe the essential pitch delivery skills required for an effective startup pitch. → presentation 8
7. Write short notes on any two: 2×5
 - a. Intellectual properties → Copyright, - -
 - b. Taxation in Nepal
 - c. Smart goal
 - d. Value proposition → why your product/service?

POKHARA ENGINEERING COLLEGE

Assessment

Level: Bachelor

Semester – Spring

Year :2025

Programme: BE Computer

Full Marks: 100

Course: [Network and Cybersecurity]

Pass Marks: 45

Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define computer security. Explain different types of attack in network security. 8
b) Define DMZ. Explain about the role of firewall in Network Security. 7
2. a) Define cryptography. Explain how transposition cipher works? 7
b) Using play fair cipher algorithm, encrypt the plain text NEPAL using the key FEDERALISM. 8
3. a) Define Substitution Cipher. Perform the encryption and decryption on the plain text INCLUDE LOVE using Vernam Cipher. 7
b) What is Discrete Logarithmic Problem? Explain the working of Diffie Hellman algorithm with example. 8
4. a) Explain the working of Elgamal Encryption with appropriate figures and explanation. 7
b) Define Hashing. Explain the generation and verification of HMAC. 8
5. a) Describe Digital Signature Algorithm and show how signing and verification is done using DSS. 8
b) What is n^2 key distribution problem? How does Kerberos solve the issue? Explain. 7
6. a) Define Web Security. Explain different layers of cyber security. 8
b) Define Transport Layer Security. Explain how SSL works. 7
7. Write short notes on: Any two
a) OWASP Top 10 5*2
b) S/MIME
c) MITM

POKHARA ENGINEERING COLLEGE
Internal Assessment Examination

Level: Bachelor
Programme: Computer 7th
Course: Engineering Economics

Year : 2026
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Items	Project A	Project B
Initial investment	Rs 200000	Rs 300000
Annual revenue	Rs 75000	Rs 85000
Annual cost	Rs 10000	Rs 12000
Useful life	6 years	8 years
Salvage value	15000	20000
MARR	10% for both projects	

$$F = P(1+i)^n$$

1. a) Define Engineering Economics. Explain its principles. 2+5
b) Define variable cost, fixed cost, sunk cost and manufacturing cost with suitable examples. 8
2. a) Mrs. Sharma is planning for her retired life and has 10 more years of service. She would like to deposit 20% of her salary, which is Rs 4000 at the end of first year and thereafter she wishes to deposit the amount with an annual increase of Rs 1000 for the next 9 years with an interest of 10%. Find the total amount at the end of 10th year. 7
b) A machine costing of Rs 500000 having estimated salvage value of Rs 50000 and useful life of 5 years. Calculate every year depreciation amount and book value under diminishing balance method and straight-line depreciation method. 8
3. a) Evaluate the following project under IRR assumption and also show unrecovered investment balance in tabular form and diagram. 7
Initial investment Rs 800000, Annual Revenue Rs 220000
Annual cost Rs 20000 Salvage value Rs 20000
MARR 10% Useful life 6 years.
b) Find both types of BC ratio and evaluate the project. 8
Initial investment Rs 300000, Annual Revenue Rs 50000
Annual cost Rs 5000 Salvage value Rs 60000
MARR 9% Useful life 10 years.
4. a) Project A and project B are mutually exclusive project. Select the best project under repeatability assumption using PW formulation. 7

- b) Calculate simple and discounted payback period of the following project. 8
Initial investment Rs 1000000, Annual Revenue Rs 300000
Annual cost Rs 20000 Salvage value Rs 20000
MARR 12% Useful life 10 years.
5. a) Perform sensitivity analysis of the following project over the range of $\pm 25\%$ in i) initial investment ii) annual revenue iii) salvage value and iii) useful life. 7
Initial investment Rs 300000, Annual Revenue Rs 55000
Salvage value Rs 60000 Useful life 10 years.
MARR 9%
Also construct the sensitivity table and plot in the sensitivity chart and find out the most sensitive parameter. 8
ABC Hydro Company gas following capital structure
Common stock Rs 1000000
Preferred stock Rs 800000
Bond Rs 2200000
Total capital Rs 4000000
common stocks were netted for Rs 985, expected dividend and growth rate were Rs 11 and 7 percent respectively.
Before tax cost of debt is 15% and tax rate is 25%
Cost of preferred stock is 12%
Calculate weighted average cost of capital of the company

$$P = 1 - \sqrt[n]{\frac{S}{B}}$$

$$LR = \frac{PW LR}{PW HR - PW LR} \times \text{diff.}$$

$$BC = \frac{PW(B)}{PW(I) - PW(CS) + PW(CO2M)}$$

(convention, direct)

Or, Define capital budgeting with its features.

6. a. What do you understand by inflation? Explain different measures of inflation. 7
- b. Suppose a company has decided to purchase a new machine having initial cost of Rs 18000 and by expecting 8-year useful life and having operating cost of Rs 3000 in 1st year. For remaining year operating cost increases per year 15% over previous year operating cost and salvage value declines each year by 20% from the previous year salvage value. Find the economic service life of this new machine when MARR is 10%. 8

Or, The annual equivalent of the defender and challenger were of in the table below. What is the best replacement strategy? where MARR=12%. The planning horizon of the project is 8 years.

Year (n)	1	2	3	4	5	6
AEC def	5300	5250	5400	5750	6200	6550
AEC chall	7700	6150	5700	5600	5675	5800

7. Write short notes on any two: 2×5
- a) Vat
 - b) Law of demand
 - c) Causes of depreciation
 - d) Common Stock

The end